In [1]:
```
<style>
div.cell {
    margin-top: 5px;
    margin-bottom: 5px;
}
</style>

import pandas as pd
import yfinance as yf
import mplfinance as mpf
```

In [5]:
```
#input parameters
ticker_symbol = 'RELIANCE.NS'     #or 'RELIANCE.NS'
start_date = '2026-01-01'
end_date ='2026-02-15'
#Download historical data
data = yf.download(ticker_symbol , start=start_date , end=end_date , auto_adjust
print(data)
```

```
[**********100%**********]  1 of 1 completed
```

Price	Adj Close	Close	High	Low	Open	\
Ticker	RELIANCE.NS	RELIANCE.NS	RELIANCE.NS	RELIANCE.NS	RELIANCE.NS	
Date						
2026-01-01	1575.599976	1575.599976	1592.500000	1571.099976	1573.699951	
2026-01-02	1592.300049	1592.300049	1594.599976	1578.199951	1580.000000	
2026-01-05	1578.099976	1578.099976	1611.800049	1575.300049	1593.000000	
2026-01-06	1507.599976	1507.599976	1569.000000	1496.300049	1569.000000	
2026-01-07	1504.199951	1504.199951	1520.000000	1498.000000	1510.000000	
2026-01-08	1470.599976	1470.599976	1503.900024	1468.800049	1500.000000	
2026-01-09	1475.300049	1475.300049	1479.900024	1465.000000	1465.000000	
2026-01-12	1483.199951	1483.199951	1485.300049	1451.000000	1475.300049	
2026-01-13	1452.800049	1452.800049	1485.800049	1444.699951	1485.000000	
2026-01-14	1458.800049	1458.800049	1467.000000	1440.199951	1444.000000	
2026-01-15	1458.800049	1458.800049	1458.800049	1458.800049	1458.800049	
2026-01-16	1457.900024	1457.900024	1480.000000	1455.099976	1458.800049	
2026-01-19	1413.599976	1413.599976	1450.599976	1403.300049	1450.599976	
2026-01-20	1394.000000	1394.000000	1416.000000	1390.000000	1410.699951	
2026-01-21	1404.599976	1404.599976	1412.900024	1373.000000	1389.500000	
2026-01-22	1402.500000	1402.500000	1419.500000	1395.000000	1410.099976	
2026-01-23	1386.099976	1386.099976	1407.000000	1381.000000	1401.800049	
2026-01-27	1380.500000	1380.500000	1391.599976	1368.000000	1380.000000	
2026-01-28	1396.699951	1396.699951	1409.900024	1384.900024	1385.000000	
2026-01-29	1391.000000	1391.000000	1402.900024	1384.000000	1402.900024	
2026-01-30	1395.400024	1395.400024	1398.000000	1378.500000	1382.599976	
2026-02-02	1390.400024	1390.400024	1392.900024	1335.500000	1340.000000	
2026-02-03	1437.099976	1437.099976	1489.500000	1430.199951	1473.900024	
2026-02-04	1456.800049	1456.800049	1464.000000	1440.300049	1444.000000	
2026-02-05	1443.400024	1443.400024	1461.500000	1439.500000	1457.000000	
2026-02-06	1450.800049	1450.800049	1452.800049	1433.500000	1441.500000	
2026-02-09	1461.599976	1461.599976	1465.900024	1453.000000	1458.000000	
2026-02-10	1458.500000	1458.500000	1471.000000	1452.199951	1471.000000	
2026-02-11	1468.699951	1468.699951	1470.000000	1453.599976	1459.599976	
2026-02-12	1448.900024	1448.900024	1473.000000	1445.500000	1470.000000	
2026-02-13	1419.599976	1419.599976	1450.699951	1416.300049	1445.500000	

Price	Volume
Ticker	RELIANCE.NS
Date	
2026-01-01	6408128
2026-01-02	6602230
2026-01-05	8704973
2026-01-06	27417125
2026-01-07	11199340
2026-01-08	16518684
2026-01-09	8335311
2026-01-12	8883745
2026-01-13	13499760
2026-01-14	8321764
2026-01-15	0
2026-01-16	17167161
2026-01-19	20392765
2026-01-20	13189498
2026-01-21	17352720
2026-01-22	15721693
2026-01-23	9837802
2026-01-27	26302396
2026-01-28	11121657
2026-01-29	20846683
2026-01-30	11239835
2026-02-02	15711165

```
2026-02-03    25456171
2026-02-04     8333387
2026-02-05    11763658
2026-02-06     8277304
2026-02-09     5775864
2026-02-10    10259630
2026-02-11     7106877
2026-02-12    11135399
2026-02-13    10755693
```

In [6]:
```
print(data.columns)
```

```
MultiIndex([('Adj Close', 'RELIANCE.NS'),
            (    'Close', 'RELIANCE.NS'),
            (     'High', 'RELIANCE.NS'),
            (      'Low', 'RELIANCE.NS'),
            (     'Open', 'RELIANCE.NS'),
            (   'Volume', 'RELIANCE.NS')],
           names=['Price', 'Ticker'])
```

In [7]:
```
#Flattened multi-level column index if present
if isinstance(data.columns , pd.MultiIndex):
    data.columns = [col[0] if isinstance(col , tuple) else col for col in data.c
print(data.columns)
```

```
Index(['Adj Close', 'Close', 'High', 'Low', 'Open', 'Volume'], dtype='object')
```

In [8]:
```
#Keep only necessary columns
ohlc_cols = ['Open','High','Low','Close','Volume']
data = data[ohlc_cols]
print(data)
```

Date	Open	High	Low	Close	Volume
2026-01-01	1573.699951	1592.500000	1571.099976	1575.599976	6408128
2026-01-02	1580.000000	1594.599976	1578.199951	1592.300049	6602230
2026-01-05	1593.000000	1611.800049	1575.300049	1578.099976	8704973
2026-01-06	1569.000000	1569.000000	1496.300049	1507.599976	27417125
2026-01-07	1510.000000	1520.000000	1498.000000	1504.199951	11199340
2026-01-08	1500.000000	1503.900024	1468.800049	1470.599976	16518684
2026-01-09	1465.000000	1479.900024	1465.000000	1475.300049	8335311
2026-01-12	1475.300049	1485.300049	1451.000000	1483.199951	8883745
2026-01-13	1485.000000	1485.800049	1444.699951	1452.800049	13499760
2026-01-14	1444.000000	1467.000000	1440.199951	1458.800049	8321764
2026-01-15	1458.800049	1458.800049	1458.800049	1458.800049	0
2026-01-16	1458.800049	1480.000000	1455.099976	1457.900024	17167161
2026-01-19	1450.599976	1450.599976	1403.300049	1413.599976	20392765
2026-01-20	1410.699951	1416.000000	1390.000000	1394.000000	13189498
2026-01-21	1389.500000	1412.900024	1373.000000	1404.599976	17352720
2026-01-22	1410.099976	1419.500000	1395.000000	1402.500000	15721693
2026-01-23	1401.800049	1407.000000	1381.000000	1386.099976	9837802
2026-01-27	1380.000000	1391.599976	1368.000000	1380.500000	26302396
2026-01-28	1385.000000	1409.900024	1384.900024	1396.699951	11121657
2026-01-29	1402.900024	1402.900024	1384.000000	1391.000000	20846683
2026-01-30	1382.599976	1398.000000	1378.500000	1395.400024	11239835
2026-02-02	1340.000000	1392.900024	1335.500000	1390.400024	15711165
2026-02-03	1473.900024	1489.500000	1430.199951	1437.099976	25456171
2026-02-04	1444.000000	1464.000000	1440.300049	1456.800049	8333387
2026-02-05	1457.000000	1461.500000	1439.500000	1443.400024	11763658
2026-02-06	1441.500000	1452.800049	1433.500000	1450.800049	8277304
2026-02-09	1458.000000	1465.900024	1453.000000	1461.599976	5775864
2026-02-10	1471.000000	1471.000000	1452.199951	1458.500000	10259630
2026-02-11	1459.599976	1470.000000	1453.599976	1468.699951	7106877
2026-02-12	1470.000000	1473.000000	1445.500000	1448.900024	11135399
2026-02-13	1445.500000	1450.699951	1416.300049	1419.599976	10755693

In [10]:
```
#Ensure all data is numeric and drop invalid rows
data = data.apply(pd.to_numeric).dropna()

print("\n Final columns dtypes:")
print(data.dtypes)
```

```
 Final columns dtypes:
Open      float64
High      float64
Low       float64
Close     float64
Volume      int64
dtype: object
```

In [11]:
```
#plot candlestick with volume
mpf.plot(
    data,
    type = 'candle',
    volume =True,
    title = f'{ticker_symbol} stock price({start_date} to {end_date})',
    ylabel = 'Price',
    ylabel_lower = 'Volume',
    style = 'yahoo',
    figratio = (14,7),
    figscale= 1.2
)
```


RELIANCE.NS stock price(2026-01-01 to 2026-02-15)
1600
1550
1500
1450
1400
1350
Price
30
20
10
0
Volume
10^6
Jan 01
Jan 08
Jan 15
Jan 22
Jan 30
Feb 06
Feb 13

In [ ]: